COMPOSITION OF THE MORPHISMS OF BIG AND SMALL ETALE SITES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [1466, 1472] and proof(s) [[1474, 1480]]; this spec covers the complete environment instead (statement 1466-1472, proof span(s) [(1474, 1480)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Editorial note. Both parts use site-level pushforward descriptions whose derivations exist in the source (the Sites-referenced construction); no load-bearing step rests on a rejected card.

Source reference. Original lemma 021H: sources/stacks-chow/tags.txt, source lines 2670–2670. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 021I: sources/stacks-chow/tags.txt, source lines 2671–2671. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Given schemes X, Y, Y in $Sch_{étale}$ and morphisms $f : X \to Y$, $g : Y \to Z$ we have $g_{big} \circ f_{big} = (g \circ f)_{big}$ and $g_{small} \circ f_{small} = (g \circ f)_{small}$.*

Proof. This follows from the simple description of pushforward and pullback for the functors on the big sites from Lemma 021H. For the functors on the small sites this follows from the description of the pushforward functors in Lemma 021I. □

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